Shaqed Carasso, Ph.D.

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PROFILE

Computational and experimental microbiologist studying how gut bacteria, bacteriophages and the host interact. My PhD showed that reversible **DNA inversions (phase variation)** in *Bacteroides* reshape bacterial function, immune modulation and long-term phage co-existence (*Cell Host & Microbe*, *Gut Microbes*, *Nature Communications*). Now extending this work to microbe–host interactions at the Broad Institute.

RESEARCH INTERESTS

Gut microbiome · Bacterial genomics and phase variation · Phage–bacteria dynamics · Microbe–host interactions · Bioinformatics

EDUCATION

2021 – 2025 **Ph.D., Bioinformatics** · Technion – Israel Institute of Technology, Haifa
Thesis: Phase Variations of the Bacteroides Genus
Advisors: Assoc. Prof. Naama Geva-Zatorsky (Technion); Dr. Itai Sharon (MIGAL Galilee Research Institute)

2018 – 2021 **M.Sc., Medical Sciences (Cum Laude)** · Technion
Thesis: Fecal Microbiota Transplantation and Phage Therapy to Eradicate Antibiotic-Resistant Bacteria
Advisor: Assoc. Prof. Naama Geva-Zatorsky

2014 – 2018 **B.Sc., Biology** · Technion

POSITIONS

08/2025 – present **Postdoctoral Associate** · Broad Institute of MIT and Harvard, Cambridge, MA
Advisor: Prof. Ramnik Xavier

2018 – 2025 **Graduate Researcher** · Technion, Haifa
Geva-Zatorsky Lab, Technion (M.Sc. and Ph.D.)

10/2021 – 03/2025 **Teaching Assistant** · Technion, Haifa
Medical Bacteriology Laboratory, undergraduate medical school course

PUBLICATIONS

Journal articles († equal contribution)

14. Ohayon C, Gross N, Capucha T, **Carasso S**, Shkedy Y, Rachmiel A, Emodi O. Efficiency and outcomes in microvascular anastomosis: a meta-analysis of mechanical versus manual techniques. *J Cranio-Maxillofac Surg*. 2025.
13. Ohayon C, Ritschl LM, Capucha T, **Carasso S**, Fichter AM, Gross N, Rachmiel A, Wolff KD, Emodi O. The never-ending story: single versus dual venous anastomoses in microsurgical radial forearm free flap for head and neck reconstruction. *J Cranio-Maxillofac Surg*. 2025.
12. †Naddaf R, †Shmilovich H, †**Carasso S**, Keshet-David R, Herren R, Gefen T, et al. Esophageal microbiome correlates with post-esophagectomy anastomotic leak in cancer patients. *ESMO Gastrointest Oncol*. 2025.
11. Gal-Mandelbaum N, **Carasso S**, Kedem A, Ziv T, Keshet-David R, Abboud R, et al. Dietary carbohydrates alter immune-modulatory functionalities and DNA inversions in *Bacteroides thetaiotaomicron*. *Nat Commun*. 2025.
10. †**Carasso S**, †Keshet-David R, Zhang J, Hajjo H, Kadosh-Kariti D, Gefen T, et al. Bacteriophage-driven DNA inversions shape bacterial functionality and long-term co-existence in *Bacteroides fragilis*. *Gut Microbes*. 2025.
9. †**Carasso S**, †Zaatry R, †Hajjo H, †Kadosh-Kariti D, Ben-Assa N, Naddaf R, et al. Inflammation and bacteriophages affect DNA inversion states and functionality of the gut microbiota. *Cell Host Microbe*. 2024.

8. †Naddaf R, †**Carasso S**, Reznick-Levi G, Hasnis E, Qarawani A, Maza I, et al. Gut microbial signatures are associated with Lynch syndrome and cancer history in Druze communities in Israel. *Sci Rep*. 2023.
7. Mandelbaum N, Zhang L, **Carasso S**, Ziv T, Lifshiz-Simon S, Davidovich I, et al. Extracellular vesicles of the Gram-positive gut symbiont *Bifidobacterium longum* induce immune-modulatory, anti-inflammatory effects. *npj Biofilms Microbiomes*. 2023.
6. †Mudrik-Zohar H, †**Carasso S**, Gefen T, Zalmanovich A, Katzir M, Cohen Y, et al. Microbiome characterization of infected diabetic foot ulcers in association with clinical outcomes: traditional cultures versus molecular sequencing methods. *Front Cell Infect Microbiol*. 2022.
5. **Carasso S**, Fishman B, Lask LS, Shochat T, Geva-Zatorsky N, Tauber E. Metagenomic analysis reveals the signature of gut microbiota associated with human chronotypes. *FASEB J*. 2021.
4. Marai I, Carasso D, **Carasso S**, Carasso S. The mechanical cost of decreasing conduction velocity: a mathematical model of pacing-induced lower strain. *J Atr Fibrillation*. 2021.
3. Berkowitz E, Kopelman Y, Kadosh D, **Carasso S**, Tiosano B, Kesler A, Geva-Zatorsky N. "More guts than brains?" The role of gut microbiota in idiopathic intracranial hypertension. *J Neuroophthalmol*. 2021.
2. †Bar-Yoseph H, †**Carasso S**, Shklar S, Korytny A, Even Dar R, Daoud H, et al. Oral capsulized fecal microbiota transplantation for eradication of carbapenemase-producing Enterobacteriaceae colonization with a metagenomic perspective. *Clin Infect Dis*. 2020.
1. Ben-Assa N, Coyne MJ, Fomenkov A, Livny J, Robins WP, Muniesa M, Carey V, **Carasso S**, et al. Analysis of a phase-variable restriction modification system of the human gut symbiont *Bacteroides fragilis*. *Nucleic Acids Res*. 2020.

Preview

P1. **Carasso S**, Hajjo H, Geva-Zatorsky N. Phage-bacteria associations: analyze, match, develop therapies. *Cell Host Microbe*. 2020.

Patent applications (pending)

1. Geva-Zatorsky N, **Carasso S**, Gefen T, Keshet R, Hajjo H, Zaatry R, Sharon I. Reversible bacterial phase-variations and uses thereof in diagnostic and therapeutic applications. WO2024176230A1, 2024.
2. Geva-Zatorsky N, Gefen T, Kariti D, Ben-Assa N, Abboud R, **Carasso S**, Zaatry R. Bacteriophages as a tool to manipulate gut commensal immune-modulation activity. WO2024194876A1, 2024.

AWARDS AND HONORS

2025	**Best Lecture Award, Technion Faculty of Medicine Graduate Research Day**
2024	**Jacobs Award for an Excellent Publication (Technion)**
2024	**Best Poster Award, Technion Faculty of Medicine Graduate Research Day**
2023	**Gutwirth Excellence Scholarship**
2022	**Teva BioInnovation Student Fellowship & Mentorship Program**
2020 – 2024	**Faculty of Medicine Excellence Scholarship for Graduate Students (five semesters)**
2017 – 2018	**Outstanding Bachelor Student Award and Dean's List (three semesters), Faculty of Biology**

CONFERENCE PRESENTATIONS

2024	**"Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages"** Israel Society for Microbiology Annual Meeting, Ramat Gan, Israel (poster)
2024	**"Long-term impact: bacteriophage-driven DNA inversions shape *Bacteroides fragilis* functionality"** 10th International Human Microbiome Consortium Congress, Rome, Italy (short talk)
2024	**"Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages"** Jacobs Graduate Research Day, Technion, Haifa, Israel (poster)

2023 **"Bacterial DNA inversions and functional plasticity during inflammation and encounter with bacteriophages"**
EMBL Symposium: The Human Microbiome, Heidelberg, Germany (talk)

2023 **"Inflammation and bacteriophages affect DNA inversion states and functionality of the gut microbiota"**
10th Annual Broad-ISF Symposium, Jerusalem, Israel (poster)

2023 **"Gut *Bacteroides* exhibit genomic phase variations under inflammatory conditions"**
ILANIT | FISEB Experimental Biology Conference, Eilat, Israel (talk)